

Research Proposal

User Acceptance of AI-Based Educational Applications Among University Students

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September 27, 2026

Contents

1	Introduction	3
2	Background of the Study	3
3	Problem Statement	3
4	Research Objectives	3
5	Research Questions	4
6	Research Hypothesis	4
7	Significance of the Study	4
8	Scope and Limitations	5
9	Literature Review	5
10	Research Methodology	5
11	Research Design	5
12	Population and Sample	6
13	Sampling Technique	6

14 Data Collection Methods	6
15 Data Analysis Techniques	6
16 Expected Outcomes	7
17 Work Plan/Timeline	7
18 References	7

1 Introduction

Artificial Intelligence (AI) has become an important technology in modern education. AI-based educational applications provide students with personalized learning, automated assistance, content generation, and interactive learning opportunities.

University students increasingly use AI-based applications for learning, programming, research, assignment preparation, and solving academic problems. However, the acceptance and effective use of these applications may depend on factors such as perceived usefulness, ease of use, trust, and students' previous experience with technology.

This study aims to investigate the factors affecting university students' acceptance of AI-based educational applications.

2 Background of the Study

AI-based educational applications include intelligent tutoring systems, AI chatbots, automated assessment tools, recommendation systems, and generative AI applications.

These applications can support students by providing quick feedback, personalized assistance, and access to learning resources. Despite these potential benefits, students may have concerns regarding accuracy, privacy, reliability, academic integrity, and dependence on AI.

Therefore, understanding students' acceptance of AI-based educational applications is important for effective implementation in higher education.

3 Problem Statement

Although AI-based educational applications are increasingly available to students, their acceptance and usage may vary among university students.

Some students may find AI applications useful and easy to use, while others may have concerns about accuracy, privacy, reliability, and academic integrity.

Therefore, this research focuses on identifying the factors that influence students' acceptance of AI-based educational applications.

4 Research Objectives

The main objective of this research is to study the acceptance of AI-based educational applications among university students.

The specific objectives are:

1. To identify the level of acceptance of AI-based educational applications among university students.
2. To examine the effect of perceived usefulness on students' acceptance of AI applications.
3. To examine the effect of perceived ease of use on students' acceptance.
4. To identify students' major concerns regarding AI-based educational applications.

5 Research Questions

1. What is the level of acceptance of AI-based educational applications among university students?
2. Does perceived usefulness influence students' acceptance of AI-based educational applications?
3. Does perceived ease of use influence students' acceptance?
4. What are the major concerns of students regarding AI-based educational applications?

6 Research Hypothesis

The following hypotheses will be tested:

- H_0 : Perceived usefulness has no significant relationship with students' acceptance of AI-based educational applications.
- H_1 : Perceived usefulness has a significant relationship with students' acceptance of AI-based educational applications.
- H_0 : Perceived ease of use has no significant relationship with students' acceptance of AI-based educational applications.
- H_1 : Perceived ease of use has a significant relationship with students' acceptance of AI-based educational applications.

7 Significance of the Study

The study may be useful to:

- University students for understanding the effective use of AI-based educational applications.
- Teachers and educators for understanding students' acceptance of AI technologies.
- Educational institutions for planning AI-supported learning activities.
- Researchers for conducting further research on AI in education.

8 Scope and Limitations

The study focuses on university students and their acceptance of AI-based educational applications.

The study is limited to selected university students and may not represent all students. The findings will also depend on the accuracy and honesty of the responses provided by participants.

9 Literature Review

Previous research on educational technology indicates that perceived usefulness and perceived ease of use are important factors in technology acceptance.

AI-based educational applications provide students with opportunities for personalized learning, automated assistance, and interactive learning. However, concerns related to privacy, reliability, accuracy, and academic integrity may affect students' willingness to use such applications.

This study will review relevant research articles to identify existing findings and research gaps related to students' acceptance of AI-based educational applications.

10 Research Methodology

This study will use a quantitative research approach. Primary data will be collected from university students using a structured questionnaire.

The questionnaire will contain demographic questions, multiple-choice questions, Yes/No questions, Likert-scale questions, rating-scale questions, and open-ended questions.

11 Research Design

A descriptive and analytical research design will be used.

The descriptive component will describe students' AI usage and acceptance, while the analytical component will examine relationships between variables such as perceived usefulness, perceived ease of use, and acceptance.

12 Population and Sample

The population of the study will consist of university students who use or have experience with AI-based educational applications.

A sample of selected university students will be taken for the study.

13 Sampling Technique

Convenience sampling will be used to select respondents who are available and willing to participate in the study.

14 Data Collection Methods

Primary data will be collected using a structured questionnaire.

The questionnaire will be distributed to selected university students through online or physical forms. Respondents will be informed that the collected information will be used only for academic research purposes.

15 Data Analysis Techniques

The collected data will be analyzed using statistical software such as Python, R, SPSS, or other relevant tools.

The following techniques may be used:

- Frequency and percentage
- Mean and standard deviation
- Correlation analysis
- Chi-square test
- Appropriate hypothesis testing

The results will be presented using tables and appropriate charts.

16 Expected Outcomes

The study is expected to identify the level of acceptance of AI-based educational applications among university students.

It is also expected to identify important factors such as perceived usefulness and perceived ease of use that may be associated with students' acceptance.

The study may also identify common concerns and challenges experienced by students while using AI-based educational applications.

17 Work Plan/Timeline

Activity	Duration	Status
Topic Selection	Week 1	Planned
Literature Review	Week 1–2	Planned
Questionnaire Design	Week 2	Planned
Pilot Survey	Week 3	Planned
Data Collection	Week 3–4	Planned
Data Analysis	Week 5	Planned
Report Writing	Week 6	Planned
Final Submission	Week 7	Planned

Table 1: Research Work Plan

18 References

1. Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*.
2. Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*.
3. Relevant research articles on AI-based educational applications, technology acceptance, and AI in higher education will be reviewed during the final research study.